



Enhancing network security using knowledge graphs and large language models for explainable threat detection

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ABSTRACT

Ensuring robust cybersecurity in modern network environments is increasingly challenging due to the growing complexity and volume of network traffic data. Traditional detection systems often fail to identify stealthy and sophisticated attacks, such as Distributed Denial of Service (DDoS), ARP poisoning, and reconnaissance scans. Moreover, many existing methods lack transparency and produce reports that are difficult for analysts to interpret, slowing both threat comprehension and response. This paper addresses these challenges by introducing a novel methodology that integrates Knowledge Graphs, XAI techniques and Large Language Models (LLMs) to enhance network threat detection, classification, explainability, and automated reporting. The proposed approach employs Graph-BERT to encode complex communication patterns and semantic relationships into enriched knowledge graphs constructed from network logs. To ensure model transparency and interpretability, Local Interpretable Model-Agnostic Explanations (LIME) are incorporated, while structured prompts guide report generation using Generative AI. Experimental results obtained on benchmark datasets demonstrate that the methodology achieves a classification accuracy exceeding 84 %, outperforming existing detection techniques. Additionally, a comprehensive evaluation involving ablation analysis, LLM-based assessments, and expert reviews shows that incorporating structured knowledge and explainability significantly enhances the clarity, correctness, and informativeness of generated reports. These findings confirm the system's effectiveness both as a detection mechanism and as a practical tool that helps analysts understand threats and craft informed responses.

1. Introduction

As computing networks continue to expand across critical sectors—including healthcare, finance, and industrial systems—the need to secure interconnected infrastructures has become increasingly urgent [1]. The growing volume of real-time data in these environments demands analytical solutions capable of swiftly and accurately detecting security threats [2]. Traditional defense mechanisms often prove inadequate due to limited resources, complex protocols, and the heterogeneity of network traffic. As a result, modern infrastructures, particularly those composed of diverse and resource-constrained devices, remain highly susceptible to advanced cyber threats, including distributed denial of service (DDoS) attacks, ARP poisoning, and man-in-the-middle attacks [3,4]. These threats affect not only individual systems but also broader network ecosystems, highlighting the urgent need for scalable, adaptive, and context-aware cybersecurity solutions [5].

To address these evolving threats, a range of cybersecurity strategies has been developed. Signature-based intrusion detection systems (IDS) and firewalls remain widely used, relying on known threat patterns to block malicious activity [6,7]. However, they are often ineffective against zero-day exploits and previously unseen attack vectors. In response, anomaly-based detection systems, which leverage statistical models or machine learning algorithms, have gained traction for their ability to detect deviations from normal behavior and identify novel threats [8]. Nevertheless, despite their promise, these systems can suffer from high false-positive rates and require ongoing tuning to remain effective in complex and heterogeneous environments [9].

In this context, autonomous artificial intelligence (AI) agents are emerging as a promising solution, increasingly adopted in network management and security operations to rapidly respond to dynamic threats. These agents can autonomously perform cybersecurity tasks on behalf of users—such as system monitoring, anomaly detection, and threat

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response—enabling a higher degree of autonomy and proactive defense. However, most AI agents, particularly those based on large language models (LLMs), operate as black boxes, often making critical security decisions (e.g., revoking user access or blocking IP addresses) without providing clear justifications. To function effectively and responsibly, such agents require two key capabilities: (i) access to accurate, structured, and context-rich information for informed decision-making; and (ii) the ability to generate transparent, interpretable explanations to foster trust and enable human oversight [10,11].

To address these challenges, this paper introduces a novel methodology—KLAGE—that combines Knowledge Graphs (KGs), eXplainable AI (XAI) techniques, and Large Language Models (LLMs) to provide a comprehensive solution for network security monitoring, threat analysis, explainability, and automated report generation. KLAGE employs Graph-BERT [12] to dynamically model network interactions as knowledge graphs, capturing intricate communication patterns, device attributes, communication protocols, traffic behaviors, and known threat signatures. This enriched context supports accurate threat detection and enables LLMs to reason more effectively about network behavior.

To ensure transparency, KLAGE integrates Local Interpretable Model-Agnostic Explanations (LIME) [13], which highlight the key features influencing each decision. Building on this explainable foundation, structured prompting with Generative AI (GAI) is used to produce coherent, actionable reports that translate analytical insights into human-readable narratives—supporting timely and informed decisions in high-stakes security environments. Overall, KLAGE enhances clarity, reduces ambiguity, and empowers cybersecurity analysts through a unified, interpretable, and data-driven framework.

Experimental evaluations demonstrate that KLAGE outperforms existing methodologies, particularly in accuracy, efficiency, and explainability. Specifically, our approach achieved over 84% accuracy in classifying network threats, surpassing existing models by more than 5%. Additionally, we evaluated multiple LLM-based report generation strategies, including both baseline models and advanced variants. The accuracy, technical depth, and clarity of these reports were assessed through an ablation study, an automated LLM-based assessment, and a domain-expert review [14,15].

The paper is structured as follows. Section 2 reviews related work, highlighting the distinct contributions of our proposed methodology. Section 3 details the KLAGE architecture, emphasizing the integration process of knowledge graphs with Graph-BERT. Section 4 discusses the experimental results. Finally, Section 6 summarizes our contributions and suggests future directions for explainable AI in cybersecurity.

2. Related work

In recent years, numerous machine learning and deep learning approaches have been developed to detect anomalies and cyberattacks, demonstrating the effectiveness of these techniques. For instance, Shon et al. [34] introduced a hybrid system leveraging Support Vector Machines (SVMs) in both supervised and unsupervised modes. Le et al. [21] analyzed deep learning models for log anomaly detection, emphasizing the critical role of pre-processing and semantic representation in improving accuracy. Fu et al. [24] explored the impact of log parsers on detection performance, noting that parsing errors can either enhance or degrade results. In the domain of IoT networks, Churcher et al. [16] compared machine learning algorithms like K-Nearest Neighbors (KNN) and Random Forest (RF) for DDoS attack classification. Their findings revealed that RF is effective for binary classification, while KNN performs better in multi-class scenarios. Additionally, Shafiq et al. [35] proposed CorrAUC, a feature selection algorithm that combines AUC and correlation metrics, significantly enhancing the detection of botnets and DDoS attacks.

2.1. Graph-based learning for network traffic

Building on the advancements in machine learning and deep learning techniques for anomaly and cyber attack detection, a recent research field has focused on leveraging graph-based methodologies to enhance network security. While traditional approaches have demonstrated significant potential in processing and analyzing structured data, the growing complexity of network traffic and the interconnected nature of cyber threats have necessitated more sophisticated tools. This has led to the emergence of Graph Neural Networks (GNNs) as a pivotal technology, offering unique advantages in modeling and analyzing network structures.

One example is the work of Jianping et al. [27], who introduced a novel approach for network attack detection through attention-based GNNs within a Federated Learning (FL) framework. Their Federated Graph Attention Network (FedGAT) enhances data privacy by training models locally on devices, avoiding the need to transfer raw data to a central server. Similarly, Boyaci et al. [18] applied GNNs to detect stealth attacks in smart grids. Their method utilizes the graph structure of the power grid and spatial correlations in measurement data. By combining model-based and data-driven approaches, they showed how GNNs can enhance the detection of sophisticated attacks, improving the security of energy management systems.

Although GNNs are powerful tools for intrusion detection and network security, they are vulnerable to adversarial attacks that can exploit their weaknesses and significantly reduce their accuracy. Zhou et al. [19] demonstrated this by developing a shadow model to craft adversarial examples targeting GNN-based systems like GCN and JK-Net. These attacks proved effective even in black-box scenarios, where attackers had limited knowledge of the system, highlighting the critical risks and limitations of GNNs in adversarial environments.

Other studies have explored the use of graph-based methods for detecting encrypted malicious traffic, highlighting their advantages over traditional detection techniques [22,25]. Advancing this field further, Yang et al. [28] introduced a framework that combines GNNs and Transformers to identify threats in encrypted traffic without relying on packet payloads, offering a robust solution for privacy-preserving traffic analysis.

Furthermore, some research has proposed transformer-based approaches that integrate advanced modeling techniques to capture temporal and structural relationships, aiming to improve anomaly and attack detection. For instance, Liu et al. [20] proposed a transformer-based framework for anomaly detection on dynamic graphs, which integrates global, local, and temporal information through a novel node encoding method and employs an edge-based sampling strategy. Similarly, Li et al. [29] presented a framework for real-time attack detection and prediction, namely DeepAG, which leverages transformers for log vectorization and a bi-directional LSTM for anomaly localization to effectively detect attacks and model non-linear dependencies using system logs and attack graphs.

2.2. Large language models for network security

In the field of security, the application of advanced language models, such as BERT-based architectures and other LLMs, is proving to be a powerful solution for addressing the growing challenges posed by these interconnected networks. These models have demonstrated remarkable capabilities in parsing, interpreting, and classifying heterogeneous security data, including system logs, network flows, and vulnerability descriptions. However, the use of LLMs in cybersecurity is still limited by a lack of contextual awareness, explainability, and reliability in several critical decision-making scenarios [36–38].

Ferrag et al. [33] introduced SecurityBERT, a model specifically designed for cyber threat detection in IoT environments. By using advanced privacy-preserving encoding techniques, SecurityBERT efficiently represents network traffic while safeguarding user privacy,

making it ideal for resource-constrained devices where performance, efficiency, and privacy are crucial for real-time security. A similar approach has been proposed in medical domains through the development of privacy-preserving and explainable federated frameworks, illustrating how LLMs can be effectively integrated with explainable AI and federated architectures to support critical decision-making while ensuring data confidentiality [39–41]. In addition, Ali et al. [32] explored the integration of BERT with Multi-Layer Perceptron for network intrusion detection, addressing class imbalance in network traffic by applying synthetic data oversampling techniques such as SMOTE [42]. Seyyar et al. [23] employed BERT to detect web-based threats such as SQL injection and cross-site scripting. By converting HTTP requests into vector representations, their method facilitated the identification of malicious activities, demonstrating the versatility of transformer-based models across diverse security contexts.

Other recent advancements in the field include the use of generative artificial intelligence to enhance vulnerability assessment. For instance, GPT-3.5 Turbo has been utilized to generate structured vulnerability descriptions, enhancing input quality for CVSS (Common Vulnerability Scoring System) scoring, a standardized framework for assessing software vulnerability severity [43]. Further exploring the role of language models, Karlsen et al. [30] evaluated models such as BERT, RoBERTa, and GPT variants for system and application log analysis, revealing their strengths and limitations in multilingual and heterogeneous settings. Ali and Kostakos [44] introduced HuntGPT, an intrusion detection system that combines machine learning-based anomaly detection with interpretability frameworks like LIME and SHAP [45,46]. Although these models offer significant advancements in cybersecurity, Bertero et al. [47] and Gupta et al. [48] demonstrated that NLP-based methods can serve both defensive cybersecurity efforts and the development of advanced cyberattacks.

2.3. Contributions of our work

The methodology proposed in this study is specifically designed to assist network analysts in identifying, interpreting, and responding effectively to sophisticated security threats targeting connected devices. Most existing approaches are limited in jointly capturing structural dependencies, semantic information, and interpretability of results, often addressing these aspects in isolation. While AI-based techniques often achieve high accuracy in identifying security threats, they typically lack transparency regarding the rationale behind their predictions, making them less useful in supporting security decision-making.

To overcome these limitations, our work proposes an integrated approach that combines graph-based analytics, advanced language models, and explainable artificial intelligence (XAI) techniques. While each individual component, such as KGs, LLMs, and XAI, has been explored in prior literature, our framework proposes a novel integration of these elements to support accurate, interpretable, and automated threat detection and reporting.

In particular, we highlight the following aspects of novelty that characterize our proposal:

- The introduction of a structured knowledge graph representation for network traffic, based on Graph-BERT, which goes beyond traditional flow analysis by incorporating semantic, structural, and temporal features to support advanced reasoning.
- The definition of a post-classification graph pruning mechanism, based on a multi-criteria significance score, specifically designed to enhance interpretability and reduce redundancy in the generated reports.
- The implementation of a prompt-based report generation component, where insights from LIME explanations are integrated into prompts for LLMs, enabling the generation of accurate and human-readable reports. By leveraging local explanations provided by LIME

alongside the classification output, KLAGe effectively addresses the trade-off between explainability and accuracy [49].

- A comprehensive evaluation strategy, including an ablation study, LLM-based qualitative and domain-specific assessments (LLM-as-a-judge), and a manual expert review, all demonstrating the benefits of integrating structured knowledge and explainability into the threat analysis pipeline.

To clearly situate our contributions within the current state-of-the-art, Table 1 summarizes recent developments in log analysis and network security, highlighting key dimensions such as employed technologies, application domains, and targeted attack types. Distinctively, our approach differentiates itself by employing structured knowledge representation and LLM-driven analysis, providing deeper insights into the interrelationships among security events.

3. Proposed methodology

This section presents an in-depth overview of KLAGe, which leverages Graph-BERT [12] to represent network nodes and their communications as a unified graph, capturing complex relationships between nodes and enabling accurate threat classification. KLAGe improves the interpretability of its classification model using LIME to identify key features in threat classification. Since raw LIME outputs can be difficult to interpret, KLAGe employs an LLM to generate structured, human-readable summaries, making threat assessments more accessible. Additionally, KLAGe enhances report generation by removing irrelevant graph nodes and summarizing network threats, ensuring administrators receive only relevant information for decision-making. The methodology follows four interconnected phases, as illustrated in Fig. 1:

- Log data is collected from devices, aggregated into traffic flows, and transformed into a set of knowledge graphs that represent individual network interactions; these individual knowledge graphs are then merged into a unified knowledge graph, which serves as the input for threat classification using Graph-BERT.
- A Graph-BERT model is used to classify the unified knowledge graph, producing in output a classified unified knowledge graph, which is subsequently interpreted through a post-hoc explainability technique based on LIME.
- The classified unified knowledge graph is pruned by removing irrelevant nodes, with the goal of reducing structural complexity and improving the interpretability of the resulting explanations.
- Finally, detailed security reports are automatically generated by combining insights from the pruned classified graph and the LIME-based explanations, enabling concise and human-readable summaries of detected threats and anomalies.

Each of these phases is discussed in the following subsections, where the corresponding technical details and design choices are presented.

3.1. Log aggregation and knowledge graph construction

Network traffic logs are typically collected using specific tools, such as Wireshark or tcpdump, which record packet traces and saves them as PCAP files. Using aggregation procedures or dedicated tools such as NetFlow, CICFlowMeter, or sFlow, these PCAP files can then be processed and aggregated into traffic flows (i.e., connections) by grouping all packets that share the same 5-tuple (source IP, destination IP, source port, destination port, and protocol). Key features, such as packet count and flow duration, are then calculated and used to classify each traffic flow as either *normal* or *attack*.

Each traffic flow is converted into a *knowledge graph* (KG), capturing enriched contextual details of network interactions [50,51]. Formally, each KG is defined as $G = (N, E)$, where N represents the set of network entities (or nodes), each identified by a port, and E denotes the set of edges representing relationships between these entities. A specific entity

Table 1

Overview of related work network security (see Refs. [16–33]).

Related Work	Technology Used	Data Type	App Domain	[L]Learn Model	Attack Type	Protocol(s)
Churcher et al. [16]	KNN, SVM, DT, NB, RF, ANN	IoT Traffic	📶	👤	DDoS, Data Exfil	↔️🌐🔒
Ioannou et al. [17]	C-SVM, OC-SVM, RBF	IoT Traffic	📶	👤👤	Select Fwd, Sinkhole	–
Boyaci et al. [18]	GNN	Smart Grid Data	⚡	👤	FDIA	–
Zhou et al. [19]	GNN, RWR, Sal Map, GCN, JK-Net	IoT Traffic	📶	👤	Net Intrusion	–
Liu et al. [20]	Transformer, ST Encoding	Dyn Graph Data	🔒	👤👤	Anom Edges	–
Le et al. [21]	DeepLog, LogAnom, CNN, RNN	Sys Logs	💻	👤	Sys Anoms	–
Fu et al. [22]	ST-Graph, RWR, Graph Rep Learn, RF	Encr Traffic (TLS)	🔒	👤	Malware, Botnet	↔️🔒
Seyyar et al. [23]	BERT, MLP	HTTP/HTTPS Req	🌐🔒	👤👤	SQLI, XSS	🌐🔒
Fu et al. [24]	ML, DL	Sys Logs	⚙️	👤👤	Anoms	–
Hong et al. [25]	GraphSAGE, KNN, TLS Traffic	Encr Traffic	🔒	👤	Malware	↔️🔒
Gupta et al. [26]	GenAI, ChatGPT, BERT, GPT	Text, Log, Net Traffic	⚙️	👤👤	Phishing, SE	–
Jianping et al. [27]	GNN, Fed Learn, Attn, GAT	Net Traffic Logs	🔒	👤	DoS, Probe	↔️🔒
Yang et al. [28]	Transformer, GNN	Encr Traffic	🔒	👤	Ransomware	↔️🔒
Li et al. [29]	LSTM	Sys Logs	🔒📊	👤	–	–
Karlsen et al. [30]	Graph Paths BERT, RoBERTa, GPT-2	Sys + App Logs	🔒📊	👤	Anom Log	–
Cui et al. [31]	LLMs (BERT, GPT-4, RoBERTa)	Sys + App Logs	🔒📊	👤👤	Events Anom Log	–
Ali et al. [32]	BERT, MLP, SMOTE	Logs	⚙️	👤👤	Events	–
Ferrag et al. [33]	BERT, BBPE, PPFL	Net Traffic	🔒	👤	DoS, Brute Force	↔️🌐
KLAGE	BERT, Graph-Bert, GPT-4, LIME	Net Traffic	📶	👤	DDoS, Ransomware	↔️🔒
		IoT/Net Logs	📶🔒	👤	DDoS, ARP Poisoning, ...	↔️🔒

App Domain: 🔒 Network Sec, 📶 IoT Sec, ⚡ Smart Grids, 💻 Soft Sys, 🌐🔒 Web Sec (HTTP/HTTPS), ⚙️ Cybersec/AIOps/Log Analysis, 📊 Sys Mon. **Learning Model:** 👤 Supervised, 👤👤 Unsupervised. **Protocols:** ↔ TCP, 🔒 UDP, 🌐 HTTP, 🔒 TLS/HTTPS.

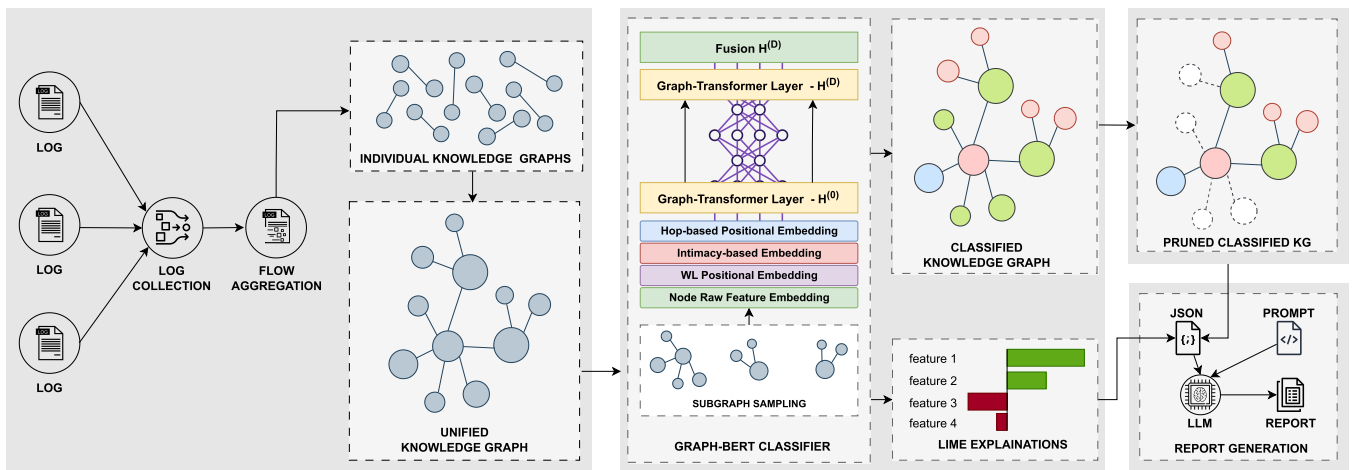


Fig. 1. Workflow of the proposed KLAGE system for enhanced network security. KLAGE integrates knowledge graph construction, graph-based threat classification, explainability, and automated report generation using large language models. The four main stages of the system are: (i) transformation of traffic logs into semantically enriched knowledge graphs; (ii) threat classification using Graph-BERT and explanation via LIME; (iii) graph pruning to focus on relevant events; and (iv) generation of human-readable threat reports using LLMs.

in the graph, denoted as N_{attack} , acts as an observer that monitors the network status to detect a particular *attack type* (e.g., ARP Poisoning, NMAP OS Detection). Prior to the identification of an attack, this entity remains isolated from the rest of the graph; however, once the attack is detected, it is linked to the corresponding attacker node [52,53]. This design enables the representation of two essential relationships within the graph:

- $E_{caused} = (N_{attack}, N_{source})$, linking a detected attack to its originating node (i.e., attacker).
- $E_{connected} = (N_{source}, N_{dest})$, representing a communication flow between a source node and a destination node.

Unlike traditional graph representations, KG representation embeds semantic information directly into the graph, enriching both nodes and edges with contextual attributes. Each node and edge in the graph is associated with a set of properties, denoted as P , which captures node-level attributes (e.g., number of packets sent, data volume, attack characteristics) and edge-level attributes (e.g., protocol type, connection duration, packet frequency).

After collecting and transforming log data into a collection of knowledge graphs, the individual graphs are unified into a single comprehensive graph. This unified representation is then used in the following step, which involves threat classification with Graph-BERT. The unification process consists of merging the log-based graphs based on communication flows between devices, resulting in a cohesive view of the network.

3.2. Threat classification with graph-BERT and post-hoc explainability via LIME

Given the unified knowledge graph obtained in the previous step, Graph-BERT enriches nodes with *Hop-based Positional Embeddings* and *Intimacy-based Embeddings*, which respectively capture the distance between nodes and the intensity of their interaction. In particular, when generating the hop-based positional embeddings, the Weisfeiler-Lehman algorithm [54] is used to assign each node a code that defines their structural role in the graph. Specifically, nodes with the same role are assigned to the same code. This code is then mapped to a vector embedding, namely the Weisfeiler-Lehman (WL) positional embedding, which encodes the structural role of a node in a graph.

The initial embedding of the i th node, h_i , is defined as follows:

$$h_i^{(0)} = f(x_i, e_i^{WL}, e_i^{hop}, e_i^{intimacy}) \quad (1)$$

where f is an aggregation function, x_i represents the features of i th node, e_i^{WL} is the Weisfeiler-Lehman positional embedding, e_i^{hop} represents the hop-based embedding between nodes, and $e_i^{intimacy}$ quantifies the closeness between nodes.

Graph-BERT processes node embeddings through a series of *graph transformer layers*, using attention mechanisms to model the relationships between nodes:

$$H^{(l)} = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V + \text{G-Res}(H^{(l-1)}, X) \quad (2)$$

where:

- $H^{(l)} \in \mathbb{R}^{n \times d}$ denotes the matrix of node embeddings at layer l , where n is the number of nodes and d is the embedding dimension;
- $X \in \mathbb{R}^{n \times f}$ denotes the initial input features of the nodes, with f being the feature dimension;
- Q , K , and V are the query, key, and value matrices for attention, obtained by linearly projecting the node embeddings from the previous layer;
- and G-Res is the residual term added to improve model stability.

The final output of the model is a fused representation of the target i th node:

$$z_i = \text{Fusion}(H^{(D)}) \quad (3)$$

which is then used for classification tasks. D denotes the total number of graph transformer layers applied in Graph-BERT, meaning $H^{(D)}$ contains the final node embeddings after the complete encoding process.

Graph-BERT does not process the entire graph at once due to the challenges of scalability and the over-smoothing problem common in traditional GNNs. To this end, it uses a *subgraph sampling* approach to generate node embeddings, where each subgraph is processed *independently*, enabling parallel training and avoiding the computational bottlenecks associated with large-scale graphs. In addition, this approach allows Graph-BERT to capture localized structural and semantic relationships while preserving the efficiency of Transformer-based learning [55,56]. BERT-Graph partitions the graph into subgraphs $G_i = (N_i, E_i)$, each centered around a node n_i and its k most similar neighbors that is determined using on the PageRank algorithm as centrality function:

$$k = k_{\min} + (k_{\max} - k_{\min}) \cdot \text{centrality}(n_i) \quad (4)$$

where $k_{\max}$ and $k_{\min}$ are hyperparameters [57,58].

Once the node embeddings are learned through the *Graph-Transformer Layers*, they are aggregated and used for classification, ensuring that the model captures both local context and global graph information without being constrained by static neighborhood aggregation. The model is subsequently fine-tuned to enhance its capability to detect multiple types of attacks. In particular, a classifier composed of fully connected layers transforms the embeddings into class probabilities, as expressed by:

$$p = \sigma(W_d \cdot \text{Dropout}(W_e \cdot h + b_e) + b_d) \quad (5)$$

where:

- h is the input node embedding,
- W_e and b_e are the weights and bias of the embedding transformation layer,
- $\text{Dropout}(\cdot)$ is a regularization function applied to prevent overfitting,
- W_d and b_d are the weights and bias of the final linear layer,
- $\sigma(\cdot)$ is the *softmax* function, which converts logits into class probabilities.

Fine-tuning is performed using the Cross-Entropy loss function:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}) \quad (6)$$

where N is the number of nodes, C number of classes (attack types) the model can predict, $y_{i,c}$ is the true label and $\hat{y}_{i,c}$ is the predicted probability for class c . Optimization is carried out using the Adam optimizer with an initial learning rate of 3×10^{-5} , adjusted dynamically.

After the Graph-BERT model is fine-tuned for inference, it classifies nodes in the unified knowledge graph and supports post-hoc explainability using LIME. Specifically, LIME analyzes the feature vectors and model predictions by performing input perturbations and observing how predictions vary. Based on this, it constructs a simple, interpretable model that highlights the most influential features contributing to each classification. The resulting explanations are formatted in JSON and passed to an LLM for generating a human-readable report. Unlike other XAI techniques such as SHAP or Integrated Gradients, LIME is a locally linear surrogate method that approximates the model's decision boundary around specific instances. While this approach may not fully capture the model's global behavior, it aligns with KLAGE's objective of providing instance-level, human-interpretable explanations that support real-time threat interpretation and automated reporting.

At the end of this step, a unified knowledge graph containing node-level threat classifications is produced, together with corresponding LIME-based feature explanations. These outputs are used as inputs for the subsequent graph pruning and report generation phases.

3.3. Post-classification graph pruning for enhanced reporting

After the inference phase, the classified unified knowledge graph is subjected to a pruning process aimed at enhancing the report genera-

tion stage by removing less relevant nodes. This process has a dual objective: (i) reducing computational complexity by simplifying the graph structure; and (ii) improving report quality by retaining only the most relevant nodes for analysis.

To determine which nodes should be removed, a *significance score* $S(n_i)$ is assigned to each node n_i in the classified graph, considering multiple structural and semantic factors. Specifically, this score is defined as:

$$S(n_i) = \alpha C(n_i) + \beta \text{sim}(n_i) + \gamma I(n_i), \quad (7)$$

where:

- $C(n_i)$ denotes the *centrality score* of node n_i , capturing its structural importance within the graph. This can be computed using degree and betweenness centrality, or alternative graph-theoretic metrics.
- $\text{sim}(n_i)$ represents the *similarity score* of n_i , measuring its average similarity to neighboring nodes based on feature-based similarity metrics.
- $I(n_i)$ quantifies the *interaction intensity* of n_i , considering the frequency and volume of data exchanged with other nodes.
- The hyperparameters α, β, γ control the relative contribution of these components.

A node n_i is pruned if $S(n_i) > \theta$, where the θ is a threshold dynamically determined as follows:

$$\theta = \mu_S + \lambda \sigma_S, \quad (8)$$

where μ_S and σ_S are the mean and standard deviation of the significance scores across all nodes, and λ is a hyperparameter regulating the pruning aggressiveness. A higher value of λ results in a more aggressive pruning strategy, discarding a larger number of low-significance nodes.

As a result, we produce a pruned and structurally simplified graph focused on high-relevance nodes, reducing input noise and guiding the LLM's attention towards the most critical aspects of the threat landscape. This approach enhances the quality and efficiency of the generated reports, as also supported by recent findings on context compression for retrieval-augmented generation [59].

3.4. LLM-driven threat report generation from pruned graphs and LIME explanations

Once the graph has been enriched through classification and pruned, it is transformed into a structured format, such as JSON, which serves as input to an LLM to generate a detailed and understandable report on the network security status. To enhance the interpretability of the classification results, this structured JSON is then extended by incorporating the explanations generated by LIME, which assesses the contribution of each feature to the model predictions. In particular, for each classified instance, LIME identifies the most influential features by introducing controlled perturbations to the input data and analyzing the resulting changes in the predicted class probability. This process can be formally described as:

$$\text{Explanation}(x) = \sum_{i=1}^k \phi_i \cdot \delta(x_i), \quad (9)$$

where x represents the input instance, k is the total number of features in the data, ϕ_i the importance weight of feature i , and $\delta(x_i)$ the feature value.

The LLM is then instructed to process the combined JSON using a custom prompt, designed to guide its interpretation. In this study, we leveraged ChatGPT-4o to generate structured reports, but the approach is generic and thus extensible to other LLMs. In particular, the LLM is instructed to act as a cybersecurity expert, processing the provided JSON for generating a report discussing detected anomalies or attacks. The following prompt has been defined to guide ChatGPT in this process, which

Let D be a dataset represented as a set of instances $D = \{x_1, \dots, x_n\}$, where each instance x_i is a vector of attributes $x_i = (a_1, \dots, a_m)$. The attributes belong to different categories:

Communication Parameters: a set $C = \{c_1, \dots, c_n\}$, including source and destination ports, transport protocol, and service.

Traffic Metrics: a set $T = \{t_1, \dots, t_k\}$, including flow duration, packets sent/received, transmitted data volume, transmission rates, inter-packet times, and average payload sizes.

Attack Label: a categorical variable y where $y \in Y$, with Y representing the attack types (e.g., ARP Poisoning, DDoS Slowloris, Brute Force SSH).

Explainability Criteria: a mapping function $E : D \rightarrow \mathbb{R}^k$, assigning each instance $x \in D$ to a k -dimensional interpretability space, derived from LIME.

Similarity Metrics: a function $S : D \times D \rightarrow \mathbb{R}$, measuring the similarity between instances based on distance metrics.

Objective: the analysis aims to detect and classify network anomalies and attacks through explainability criteria and correlations between network parameters. The report must determine the presence of attacks, origin, affected nodes, impact, periodicity, and service disruptions. Additionally, similarity analysis between connections should reveal behavioral clustering and structural relationships.

Expected Output: a structured report summarizing network anomalies, attack classification, correlation patterns, and similarity metrics. The report must provide data-driven insights on attack validation, propagation, periodicity, and affected services, referencing key metrics.

Communication Parameters: { Source Port: 52088, Destination Port: 443, Protocol: TCP, Service: SSL }.

Traffic Metrics: { Flow Duration: 0.257152 sec, Forward Packets: 13, Backward Packets: 12, ... }.

Similarity Metrics: Connection to Node 20 detected with a similarity score of 579.59.

Attack Label: ARP Poisoning.

Explainability Criteria: { $fwd_pkts_payload.avg > 47.26 = 0.191$, $fwd_data_pkts_tot > 3.00 = -0.123$, $1.70 < fwd_pkts_per_sec \leq 1040.42 = 0.121, \dots$ }.

Objective: the analysis aims to detect and classify network anomalies and attacks through explainability criteria and correlations between network parameters. The report must determine the presence of attacks, origin, affected nodes, impact, periodicity, and service disruptions. Additionally, similarity analysis between connections should reveal behavioral clustering and structural relationships.

Expected Output: a structured report summarizing network anomalies, attack classification, correlation patterns, and similarity metrics. The report must provide data-driven insights on attack validation, propagation, periodicity, and affected services, referencing key metrics.

defines key aspects of the report for ensuring that relevant information is extracted and analyzed effectively.

Below is an example of a prompt built from the subgraph including data related to a specific attack.

By instructing LLMs in this way, KLAGÉ automates the process of transforming complex network data into actionable reports, helping users quickly interpret and respond to detected threats. These results align with prior research demonstrating that prompt-based LLMs, when

fed with structured and representative input data, can effectively generate high-quality and well-organized analytical reports [60–62].

3.5. Complexity analysis

Let $|L|$ be the number of raw packet logs and $|F|$ the number of traffic flows produced after aggregation. When these flows are converted and merged, they form a *unified knowledge graph* with $|N|$ nodes and $|E|$ edges. Graph-BERT is instantiated with H layers, A attention heads, and hidden dimension d . The post-hoc explainer performs M LIME perturbations, and the pruning step keeps only a fraction p of the nodes and edges, i.e. $|N_p| = p|N|$ and $|E_p| \approx p|E|$. The complexity of each stage is as follows:

- *Pre-processing (logs $\rightarrow$ flows $\rightarrow$ unified graph)*. Transforming logs into flows costs $O(|L|)$. Building graphs from the flows and merging them into the unified graph costs $O(|F| + |E| \log |N|)$, where the $\log |N|$ factor arises from hashing/deduplication during the merge.
- *Graph-BERT classification*. One forward pass on the unified graph costs $O(H A d (|N|^2 + |E|))$.
- *Graph pruning*. Removing irrelevant nodes and edges costs $O(|N| + |E|)$.
- *LIME explanations*. Each of the M perturbations re-runs the classifier, adding $O(M H A d (|N|^2 + |E|))$.
- *Prompt construction for report generation*. Preparing the input for the LLM, by serializing the pruned graph and integrating LIME explanations, is linear in the pruned graph size, costing $O(|N_p| + |E_p|) = O(p(|N| + |E|))$.

Considering the dominant terms, the overall complexity is dominated by the Graph-BERT classification and LIME explanation stages, with total cost $O(M H A d (|N|^2 + |E|))$.

4. Experimental results

To assess the effectiveness of the proposed methodology, we used two public datasets: RT-IoT2022 [63] and CIC-IoT2023 [64]. These datasets contain logs describing various advanced network attack types, including DDoS, brute-force attacks, and spoofing. While RT-IoT2022 is based on a real-time industrial IoT infrastructure, CIC-IoT2023 focuses on a home IoT environment. Both datasets share common features based on NetFlow, a network protocol originally developed by Cisco to monitor IP traffic by aggregating packets into traffic flows. Specifically, NetFlow-based features include source and destination ports for identifying communication endpoints, protocols (e.g., TCP, UDP, ICMP), services (e.g., DNS, HTTP, SSL), flow duration, and packet traffic metrics (e.g., packets sent and received). Experiments were conducted on a standard GPU-enabled system featuring a 96-core Intel Xeon CPU (3.00 GHz) and an NVIDIA A30 GPU with 24 GB of memory. On average, Graph-BERT required approximately 39 s per training epoch, resulting in a total training time of just over 1 h for 100 epochs. During inference, it achieved a latency of around 90 ms per batch (256 samples) and a throughput exceeding 2600 samples per second, utilizing approximately 10 GB of GPU memory.

Table 2 presents a sample of traffic flow features from the RT-IoT2022 dataset. It is important to note that IP addresses are not present

in the table, since in the dataset they were removed to prevent overfitting during the training model. However, when classifying flows in a real-world context, preserving IP information can enrich reports with details about the devices involved (for example, identifying the source and target of an attack), facilitate attacker profiling, and support effective incident response.

The graph in Fig. 2(a) visualizes 1000 IoT network traffic flows, showing malicious interactions. Red nodes represent source ports, green nodes represent destination ports, and blue nodes indicate attack types. Each port is treated as a node in the network. As such, red nodes are interpreted as attackers (via their source ports), while green nodes represent the victims (via their destination ports). Yellow edges illustrate the *caused* relationship between an attack type and its originating source port, while orange edges indicate connections between source ports (attackers) and destination ports (victims). Fig. 2(b) visualizes malicious network activities by mapping the relationships among attack types, source ports, and target ports. In the graph, blue nodes denote distinct attacks, i.e., *DDoS Slowloris*, *OS Detection*, and *TCP Port Scan*. Red nodes correspond to source ports utilized by potentially malicious devices, while green nodes highlight frequently targeted destination ports, including port 80 (commonly used for exposing HTTP applications) and port 1443.

The attack labeled DDoS Slowloris is associated with several source ports such as 55 294, 55 182, 55 076, and 55,382 which all target port 80, indicating a distributed nature. The TCP Port Scan attack is linked to source port 59 247, which attempts to connect to port 1443, suggesting scanning activity aimed at services on that port. Meanwhile, OS Detection originates from source port 46,600 and also targets port 1443, reflecting reconnaissance behavior aimed at fingerprinting the operating system of the destination host.

4.1. Optimal technique selection for classification

In this section, we evaluate the performance of KLAGe by comparing it against four advanced architectures recently proposed in the literature (see Section 2). The results of this comparison are summarized in Table 3, where we report the accuracy, precision, recall, and F1-score for each method in the task of cyberattack classification.

Among all the models, KLAGe achieves the highest F1-score at 84.1 %, outperforming the second-best model by over 5 %. Notably, KLAGe also leads across all other evaluation metrics, including accuracy (84.1 %) precision (83.5 %), and recall (84.7 %), indicating a balanced and consistent performance across different aspects of classification. The observed gains can be attributed to KLAGe's integration of domain-specific knowledge graphs, which enables a more structured and semantically rich representation of the underlying network data.

4.2. Report generation

This section evaluates the performance of various methods for generating security reports, with the primary goal of comparing the effectiveness of reports produced by various automated approaches. Specifically, we consider the following methods:

- **GPT-base**: this method processes raw network logs in a zero-shot setting. The language model is provided with unstructured network

Table 2
Example of network traffic flows from the RT-IoT2022 dataset.

Protocol	Service	Flow Duration [ms]	Packets Sent/Recv	Bytes Sent	Source Port	Destination Port	Attack Type
TCP	HTTP	1500	50/45	5000	53,824	80	ARP Poisoning
UDP	DNS	2000	30/25	3000	60,932	53	DDoS Slowloris
TCP	SSH	500	100/95	10,000	49,152	22	Brute Force
TCP	HTTP	1200	60/55	6000	54,011	8080	OS Detection
TCP	HTTPS	800	40/35	4000	55,321	8443	TCP Port Scan
UDP	DNS	1000	70/65	7000	61,678	5353	DoS SYN Flood

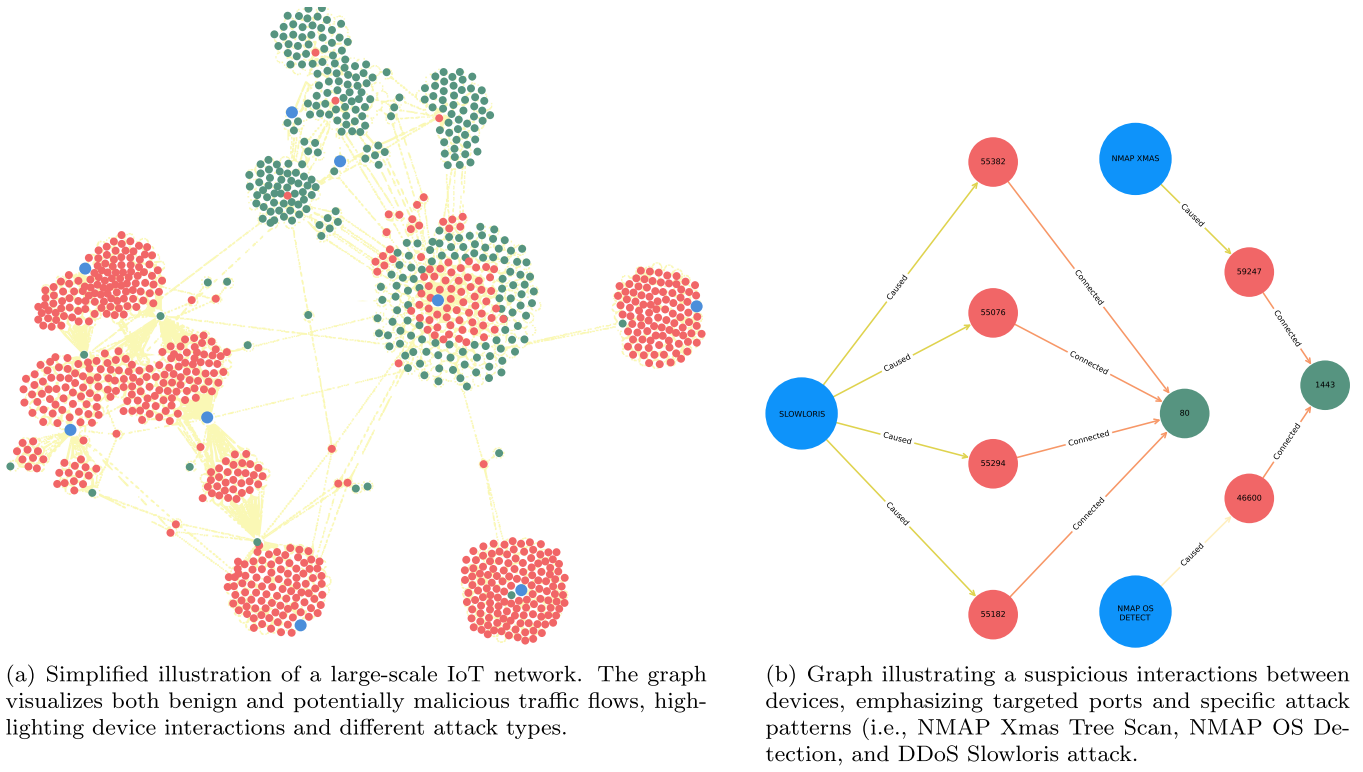


Fig. 2. Graph-based analysis of IoT network traffic flows.

Table 3
Comparison of state-of-the-art architectures for cyberattack detection.

	Technologies	Acc.	Prec.	Rec.	F1
Boyaci et al. [18]	GNN	79.1 %	78.0 %	80.0 %	79.0 %
Zou et al. [19]	GNN, RWR, Sal Map	78.6 %	77.5 %	79.0 %	78.2 %
Hong et al. [25]	GraphSAGE, KNN	79.0 %	78.2 %	79.5 %	78.8 %
Ali et al. [32]	BERT, MLP, SMOTE	79.0 %	78.5 %	79.2 %	78.8 %
KLAGE	KG, Graph-BERT, GPT-4o	84.1 %	83.5 %	84.7 %	84.1 %

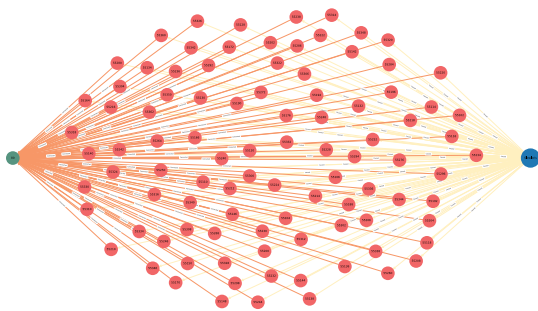
logs and is tasked with generating reports without any additional structured information or prior knowledge of node classifications.

- GPT-int (*intermediate*): this approach analyzes network logs using information about node classifications, but it does not leverage advanced relational structures, such as knowledge graphs. Nonetheless, the inclusion of class information enhances the model's ability to interpret logs, resulting in improved report generation.
- GPT-adv (*advanced*): building upon the intermediate approach, this method fully leverages the benefits of knowledge graphs. It incor-

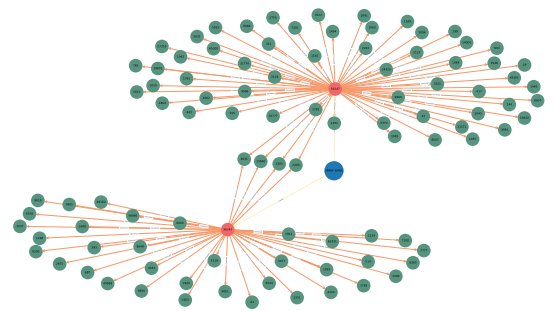
porates contextual information about the network structure, node similarities, and explanatory features (from LIME), enabling the generation of more detailed, accurate, and interpretable reports.

Two examples of reports are presented in the following subsections to illustrate the performance of the GPT-base and GPT-adv approaches on different types of attacks. The GPT-int method has been omitted for the sake of brevity.

In the next sections, an in-depth evaluation is carried out to assess the accuracy, technical depth, and overall clarity of the generated re-



(a) Representation of a DDoS Slowloris Attack from the RT-IoT2022 dataset.



(b) Representation of an NMAP XMAS Tree Scan from the CIC-IoT2023 dataset.

Fig. 3. Visual representations of specific attack types from the two datasets.

ports. Specifically, the evaluation includes: (i) an ablation analysis to examine the influence of various features on the accuracy, completeness, and overall quality of the reports (§ 4.3); (ii) the use of an LLM as an independent evaluator (i.e., LLM-as-a-judge) to assess reports based on qualitative criteria or their ability to provide answers to domain-specific aspects (§ 4.4); and (iii) a manual evaluation performed by cybersecurity experts (§ 4.5).

Example report 1: slowloris DDoS attack. Fig. 3(a) provides a representation of DDoS Slowloris activity within an IoT network, highlighting the dynamics of source and destination ports under attack, particularly the consistent targeting of port 80 (HTTP) over TCP. The figure illustrates the structured and coordinated nature of the attack, where multiple nodes act in unison to exhaust server resources by keeping numerous connections open without completing requests.

Below is an excerpt from the report generated by the language model (GPT-4o) using the GPT-adv method, based on the Slowloris attack depicted in Fig. 3. The report provides a detailed and structured analysis of the attack, highlighting specific characteristics that aid in its identification. For example, the attack uses only TCP and targets port 80, with the goal of disrupting web services by overwhelming the connection capacity of the server. Each connection involves a small number of packets with low payloads (26–47 bytes forward, 0 bytes backward) and persists for extended periods, often over 30 s, without completing requests. Traffic is intentionally slow, typically less than one packet per second, to evade rate-based detection systems, although occasional spikes in excess of 1000 packets per second are observed. The attack is characterized by high forward interarrival times and minimal backward traffic, indicating that the attacker is keeping connections open to block legitimate access. Additionally, the consistent targeting of port 80, the large number of connections, and the high correlation between multiple flows indicate a coordinated lateral movement campaign involving compromised hosts within the network.

Below is the report, generated using the GPT-base method, for the DDoS Slowloris attack discussed previously. In this zero-shot learning setup, the language model fails to identify the Slowloris attack or provide specific insights into the nature of the threat. Instead, it only offers general observations about the network logs, such as mentions of repeated connections and unusual flow durations, without recognizing the pattern as indicative of a targeted DoS technique. In many other cases, this base approach fails to detect the attack entirely or incorrectly identifies a different type of threat. Sometimes, it simply reports that the logs may be indicative of one or more possible attacks, without offering a clear classification.

Example report 2: nmap XMAS tree scan attack. Below another report, generated using the GPT-adv method, for the attack illustrated in Fig. 3(b). The report provides an in-depth analysis of network connections that are indicative of an *Nmap XMAS Tree Scan*, a high-throughput TCP port scanning technique used for reconnaissance. It effectively highlights key indicators such as single-packet bidirectional flows, missing payloads, short durations, and the use of specific TCP flags (FIN, PSF, URG) to evade detection. Multiple port targeting and extremely high throughput rates, peaking at over 800,000 packets per second, further confirm an automated and illegitimate activity. The report also notes structured scanning behavior, including repeated use of certain source ports and similarities between flows, indicating a deliberate mapping effort.

The next report has been generated for the same attack (Nmap XMAS Tree Scan), using the GPT-base reporting approach in a zero-shot setting. While it succeeds in identifying suspicious TCP behavior and reasonably outlines multiple hypotheses (e.g., the possibility of either an XMAS Tree Scan or OS detection via TCP/IP fingerprinting), it highlights several limitations typical of this basic analysis model. Notably, the report lacks decisive classification, provides only surface-level reasoning,

DDoS Slowloris Attack: Network Vulnerability Analysis

The DDoS Slowloris attack is evident across the analyzed connections, revealing a structured and persistent attempt to exhaust a web server's available connections. The attack operates exclusively over TCP, targeting port 80 (HTTP), with all interactions maintaining a low and sustained packet rate to remain undetected while consuming server resources. The flow durations vary significantly, with some connections persisting for over 30 s, a hallmark of Slowloris-based attacks, which aim to hold connections open as long as possible without completing requests. [...]

Each connection consists of a small number of packets, with 4–8 forward packets and 3–13 backward packets per connection. The average forward payload sizes range between 26 and 47 bytes, while the backward payload remains consistently at 0 bytes, confirming that the target server is receiving but not responding with substantial data.

Packet rates remain low—often under 1 packet per second—to evade rate-based intrusion systems, though some flows spike above 1000 packets per second, likely to sustain attack pressure. High forward inter-arrival times and minimal backward traffic suggest the attacker is keeping connections alive while blocking legitimate access.

The attack vector is classified as lateral movement, suggesting this activity may be part of a larger campaign involving multiple compromised hosts within the network. The consistent targeting of port 80 confirms the objective is to disrupt web services, potentially leading to a denial of service for legitimate users. [...]

Given the structured nature of the attack and the large number of connections, there is a strong probability that compromised machines within the network are participating in a coordinated DDoS effort. [...] The strong correlation among multiple connections indicates a distributed strategy, with distinct originating hosts operating in unison to maximize the attack's impact. [...]

and fails to correlate specific technical evidence (such as TCP flag patterns) with a concrete attack signature. This uncertainty shows that the model has trouble confidently identifying stealthy or low-visibility network attacks. As a result, its conclusions are mostly general, without providing clear or actionable information about the threat.

4.3. Ablation analysis

This section presents an ablation analysis to assess how well LLMs generate accurate descriptions of network attacks from raw log data. In particular, each generated report is compared to a reference JSON that includes the actual attack name, classification elements (e.g., protocols and attack vectors), relevant network components (e.g., port numbers and flow durations), and other explanatory features. To evaluate the quality of the generated reports, four key scores have been used:

- **Attack Identification (I_A)**, which quantifies how effectively the report identifies the known attack. Scores range from 0 to 1, with higher values indicating stronger recognition of the attack.
- **Classification Consistency (C_C)**, which measures how accurately the report describes communication details such as source port, destination port, protocol, and service.
- **Network Comprehension (N_C)**, which assesses how well the report captures traffic characteristics and similarity patterns, reflecting the completeness of network descriptions.
- **Explainability (E_X)**, which evaluates the extent to which the report incorporates insights from XAI methods (e.g., LIME), ensuring that the reasoning behind model predictions is included and clearly articulated.

Table 4

Ablation analysis results for LLM-generated network attack reports. Higher scores indicate better alignment with the reference network data and explanatory features.

Attack Type	Report Approach	Attack Identification (I_A)	Classification Consistency (C_C)	Network Comprehension (N_C)	Explainability (E_X)
ARP Poisoning	base	0.61	0.32	0.21	0.27
	int	0.85	0.57	0.52	0.34
	adv	1.00	0.91	0.94	0.83
DDoS Slowloris	base	0.52	0.41	0.35	0.23
	int	0.92	0.67	0.64	0.46
	adv	1.00	0.94	0.91	0.82
DoS SYN Flood	base	0.53	0.31	0.25	0.23
	int	0.74	0.54	0.43	0.37
	adv	1.00	0.95	0.87	0.71
OS Detection	base	0.55	0.38	0.33	0.18
	int	0.83	0.51	0.58	0.35
	adv	1.00	0.96	0.92	0.77
TCP Port Scan	base	0.58	0.46	0.39	0.29
	int	0.94	0.64	0.62	0.42
	adv	1.00	0.97	0.94	0.86
Brute Force	base	0.50	0.35	0.28	0.14
	int	0.76	0.52	0.49	0.39
	adv	1.00	0.93	0.85	0.79

Unusual Network Behavior Analysis

This report provides an in-depth examination of anomalous network behavior observed in a scenario where web services become unresponsive due to a large volume of incomplete and prolonged connection attempts. The analysis explores key technical aspects such as port utilization, communication protocols, and flow duration patterns, offering insights into how such behavior can strain and degrade server performance.

The activity is consistently directed at port 80, which is commonly associated with HTTP-based web services. This consistent targeting suggests an intent to interfere specifically with web-based functionalities. On the other hand, source ports vary widely across the dataset—characteristic of high-connection-rate events, where multiple unique flows are initiated in parallel to simulate legitimate traffic.

The interactions are based entirely on the TCP protocol, known for its connection-oriented nature. This choice enables the establishment of numerous sessions that remain in a half-completed state. [...]

An evident disparity in flow durations is observed, ranging from extremely brief to significantly extended intervals. The shorter flows indicate frequent initiation attempts, whereas the longer flows align with behavior aimed at occupying server-side resources. This extended engagement without completion introduces inefficiencies in resource handling on the receiving end.

The connections exhibit a repetitive and highly uniform structure, pointing toward an automated process designed to scale and persist. Many flows show minimal variation in packet characteristics, reinforcing the notion of a systematic approach. The consistent targeting of HTTP services reveals a deliberate strategy to degrade performance at the application layer. [...]

To calculate these scores, we employed a Sentence-BERT model (all-MiniLM-L6-v2) to assess how effectively specific aspects of network traffic, extracted from the original JSON, are semantically represented in the generated report. Specifically, text segments are transformed into semantic embeddings using the approach proposed in

Reimers and Gurevych [65], and then compared using cosine similarity. This enables the detection of conceptual matches, even when the information is conveyed using different wording.

The ablation analysis was conducted by generating ten reports for each attack using the three automated approaches described earlier (GPT-base, GPT-int, and GPT-adv). Specifically, for both the RT-IoT2022 and CIC-IoT2023 datasets, we generated ten reports for each of six different attacks (ARP Poisoning, DDoS Slowloris, DoS SYN Flood, OS Detection, TCP Port Scan, and Brute Force).

Table 4 reports the average values obtained across these reports. The results indicate that the attack identification score (I_A) achieves its highest performance with the advanced approach, which consistently detects all attack types ($I_A = 1.0$). The intermediate approach shows a moderate performance increase over the GPT-base, which remains the least reliable. A similar trend is observed for the classification consistency score (C_C), where the advanced method reaches values up to $C_C = 0.97$, significantly reducing ambiguity in describing communication attributes such as ports, protocols, and services. The network comprehension score (N_C) further reinforces this trend, demonstrating that the advanced approach, enhanced with the use of knowledge graphs, excels in capturing node relationships and packet behavior ($N_C \approx 0.9$). In contrast, GPT-base reports lack depth, and the intermediate approach offers only partial improvements. Finally, the explainability score (E_X) follows a similar pattern: while the zero-shot approach (GPT-base) provides minimal justification ($E_X < 0.3$), the advanced approach produces more structured and comprehensible explanations aligned with the insights provided by LIME.

4.4. LLM-as-a-judge

The ablation study highlighted the impact of structured knowledge and classification awareness in enhancing report completeness. To further evaluate the effectiveness of the different reporting strategies, we employed an LLM as an automated evaluator (i.e., LLM-as-a-judge), which systematically assigns scores to the reports based on various critical aspects [66,67]. Specifically, the analysis consists of two complementary evaluations: (i) a *qualitative assessment*, which measures general attributes such as coherence and informativeness; and a (ii) *domain-specific assessment*, which analyzes the accuracy of security-related details. Also in this evaluation, for each dataset, we generated ten reports per attack and asked the LLM to evaluate them.

Nmap XMAS Tree Scan: Network Reconnaissance Analysis

The analysis of network connections indicates a systematic and high-speed reconnaissance operation characteristic of an Nmap XMAS Tree Scan, a form of TCP Port SCAN attack. Each connection comprises precisely one forward and one backward packet, with no legitimate data exchange occurring. The absence of payload data, coupled with flow durations of microseconds, reinforces the hypothesis that this activity is purely exploratory and aimed at mapping network services. [...]

The scan targets a variety of destination ports, including 7777, 311, 1218, 1050, and 3390, suggesting a broad effort to identify potential entry points across different services. The attack employs a structured TCP-based approach, with each packet crafted to exploit variations in how different operating systems and network devices respond to specific TCP flag combinations. Notably, the XMAS scan sets the FIN, PSH, and URG flags in the TCP header. [...]

Observed transmission rates are extraordinarily high, frequently surpassing 100,000 packets per second, with peaks reaching 838,860 packets per second. This volume of activity is far beyond typical user-initiated communication and aligns with automated scanning behavior. The lack of an associated service designation in the analyzed connections further supports the conclusion that no legitimate application-layer interactions occurred. [...]

The repeated use of specific source ports, particularly 36,242 and 59,247, suggests a predictable yet deliberate methodology, iterating through target hosts and ports efficiently. This behavior is consistent with pre-intrusion reconnaissance, where an attacker maps network resources before attempting exploitation. [...]

The absence of periodicity in the scan's execution suggests a single, high-speed probing event rather than a persistent monitoring effort. However, similarity metrics between connections indicate a structured approach, with notable correlations observed between certain flows. [...]

Defensive measures should focus on real-time intrusion detection and anomaly-based systems to flag high-speed scans. Firewall rules must ensure unnecessary ports are closed, and TCP rate-limiting should be applied to hinder reconnaissance. [...]

Ambiguous TCP Scanning Activity: XMAS Tree Scan or OS Detection?

This report provides an analysis of a set of suspicious TCP-based flows exhibiting characteristics commonly associated with reconnaissance activities. While certain elements suggest the presence of an Nmap XMAS Tree Scan, other indicators are also consistent with OS detection techniques such as TCP/IP stack fingerprinting. [...]

The data reveals scanning across a broad range of ports, including both well-known services like 80 (HTTP) and 443 (HTTPS), as well as higher ephemeral ports. This wide targeting aligns with behaviors typical of both XMAS scans—intended to identify open ports—and OS detection probes, which may target specific ports known to produce OS-specific responses. [...]

All flows are based on the TCP protocol and do not consistently correlate with any identifiable service. This could be explained by the use of custom or unusual TCP flags, suggestive of an XMAS Tree Scan, but may also result from crafted probes for OS fingerprinting that do not engage in full protocol handshakes. Both types of attacks are known for generating ambiguous service signatures, further complicating detection based solely on service mappings.

The flows are very short-lived, often lasting only a few microseconds or milliseconds. This brevity is a hallmark of non-intrusive scanning activities, which aim to minimize detection. While such short durations are consistent with XMAS scans, they are also typical of OS detection tools that send isolated probes to trigger low-level TCP/IP stack responses. The low variance in flow durations across the dataset limits their usefulness in clearly distinguishing between the two attack types. [...]

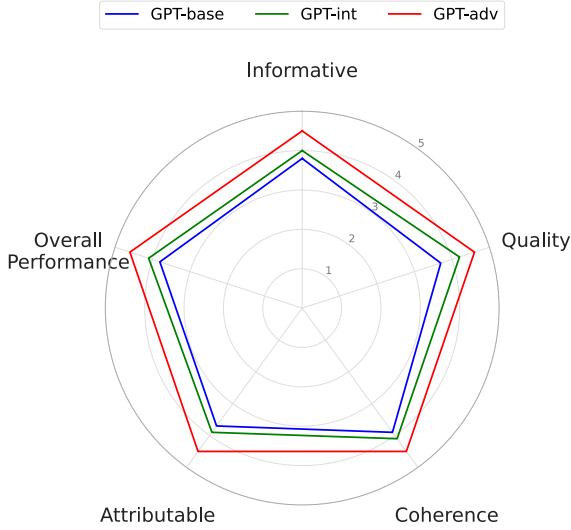
The first evaluation, the results of which are shown in Fig. 4(a), evaluated the reports using the *ChatGPT o1* model, which was tasked with assigning scores on a scale of 1 (worst) to 5 (best) based on five criteria: (i) *informativeness*, which measures the extent to which the report includes crucial details while maintaining accuracy and conciseness; (ii) *quality*, which assesses the overall clarity, coherence, and structure; (iii) *coherence*, which assesses the logical organization and readability of the report; (iv) *attributability*, which determines whether all information in the report can be traced back to the original source; and (v) *overall performance*, which reflects the ability of the report to communicate information effectively and logically. The results indicate that GPT-adv achieved the highest scores in all categories, demonstrating a significant improvement in report quality over both the GPT-base and GPT-int methods, due to its integration of structured knowledge and explanatory features.

The second evaluation, shown in Fig. 4(b), focused on domain-specific aspects related to network security, assessing six key dimensions: (i) *attack confirmation*, which determines whether the report correctly identifies the presence of an attack; (ii) *attack start time*, which evaluates the accuracy of the reported attack initiation time; (iii) *attack origin*, which assesses whether the report correctly identifies the origin of the attack; (iv) *affected nodes*, which measures the correctness of the reported impacted network nodes; (v) *periodic pattern*, which eval-

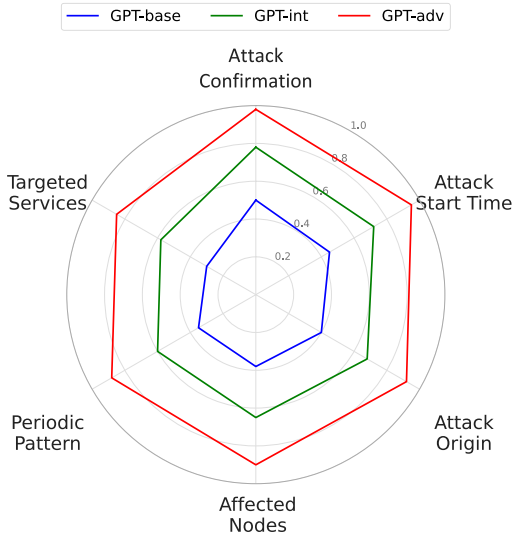
uates whether the report identifies recurring attack behaviors; and (vi) *targeted services*, which determines whether the report correctly identifies the network services targeted by the attack. In this evaluation, GPT-adv also demonstrated superior performance, achieving the highest values in identifying domain-specific details. In contrast, the GPT-base model presents difficulties, in particular in identifying attacks and affected nodes. The intermediate model, GPT-int, shows variable results: in some aspects, it obtains quite good results, close to those obtained by the advanced model, while in others it shows a drop in performance. A critical aspect that emerged from the analysis is that less advanced models, GPT-base and GPT-int, often generate inaccurate information, even when the attack is explicitly present in the input. For example, these models often incorrectly attribute attacks to non-existent nodes and generate attack patterns that were not present in the original data. This phenomenon does not occur in the advanced model, which uses structured information to improve the reliability and overall quality of the reports generated by the LLM.

4.5. Domain expert evaluation

To further validate the effectiveness of the generated reports, we conducted a direct evaluation campaign involving cybersecurity experts. Specifically, we engaged ten cybersecurity professionals with varying levels of expertise—from beginners to advanced practitioners—and asked them to assess a set of reports based on informativeness, technical detail, clarity, precision, and overall preference. Each participant completed a survey in which they reviewed the reports generated for five different attacks. Specifically, for each attack, three reports were provided, generated using the *base*, *intermediate*, and *advanced* approaches. In addition, to enable the experts to assess the correctness of each generated report, the corresponding logs used for its generation were also provided. To mitigate presentation bias, the corresponding reports for each attack



(a) Qualitative evaluation with ChatGPT based on general aspects such as coherence, clarity, and informativeness.



(b) Domain-specific evaluation with ChatGPT focused on the accuracy of security-related details.

Fig. 4. LLM-as-a-judge: comparison between general qualitative and domain-specific evaluations of the generated reports.

were presented in random order. Participants were asked to evaluate and select the best report for each attack based on the following four criteria:

- (i) Which report provides the *most comprehensive and informative content* about the attack?
- (ii) Which report demonstrates the *highest level of technical detail and expertise*?
- (iii) Which report offers the *clearest and most well-structured explanation* of the attack?
- (iv) Which report is the *most accurate and precise* in its descriptions and analysis?
- (v) Which report do you consider the *overall best* in terms of quality and usefulness?

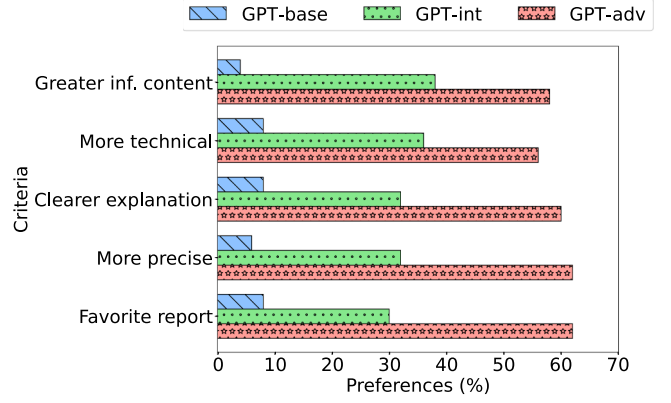


Fig. 5. Expert-based evaluation of the generated reports. Ten cybersecurity professionals assessed reports across five dimensions: informativeness, technical accuracy, clarity, precision, and overall quality.

Fig. 5 presents the results of the expert evaluation. As shown, GPT-adv consistently received the highest preference across all evaluation criteria, with on average 60 % of the experts considered it capable of generating reports with superior characteristics compared to the other approaches. In contrast, GPT-base was rated poorly across all aspects, receiving user preference in only 7 % of cases on average. Lastly, GPT-int, receiving an average of 33 % of preferences, performed better than the base model but remained significantly below the advanced approach. Overall, these results reflect the clear appreciation of experts for more advanced reports, which include comprehensive technical details and clearer information, supported by row logs, explanations, and structured knowledge aligned with attack logs.

5. Discussion and future work

The graph-based architecture of KLAGe enables the detection of suspicious patterns and supports the seamless integration of new data sources and adaptive strategies. Its evaluation against state-of-the-art approaches, emphasizing semantic modeling and LLM integration, demonstrates its suitability for evolving security scenarios and its potential to address novel and unforeseen threats.

KLAGe is designed to detect network-centric threats (e.g., DDoS, ARP poisoning, and port scans), which exhibit clear patterns suitable for coding via knowledge graphs. Advanced persistent threats (APTs) and polymorphic malware pose greater challenges due to their stealthy and adaptive behavior. These threat classes typically require a combination of longitudinal behavioral analysis, host-level telemetry, and advanced memory inspection, which are beyond the current architectural scope of KLAGe. Future work will focus on integrating temporal knowledge graphs and host-level telemetry to improve detection of these advanced threats.

Upcoming efforts will focus on enriching the knowledge graph with dynamic and heterogeneous data sources, such as contextual signals from IoT devices, edge nodes, and real-time threat intelligence feeds. We also plan to explore the integration of continual learning strategies and online adaptation mechanisms, allowing the framework to update its internal models in response to new traffic behaviors and emerging threat patterns. By doing so, KLAGe will further extend its ability to generalize across evolving attack surfaces and support real-time decision-making with higher levels of precision, transparency, and semantic awareness.

While KLAGe incorporates several mechanisms aimed at controlling complexity, such as subgraph sampling (to reduce the size of the input to Graph-BERT) and post-classification pruning (to streamline LLM input), we plan to integrate time-windowed processing pipelines that support scalable log aggregation into flows from diverse sources, involving powerful and scalable frameworks like Apache Kafka (for high-throughput

ingestion of log data from heterogeneous sources) and Apache Flink or Storm (for real-time stream processing and windowed aggregations). By operating over sliding or tumbling time windows, KLAGÉ will be able to capture evolving network behaviors in near real time, facilitating timely detection of anomalous activity and supporting the generation of contextualized knowledge graphs that reflect the dynamic nature of network environments. A reference implementation would involve using nProbe to export NetFlow data into Kafka topics, processed in Flink to perform flow-level filtering, enrichment with threat intelligence feeds, and batching into JSON-formatted inputs for the KLAGÉ classification pipeline. Additionally, the integration with Grafana or Elastic Stack would enable real-time monitoring and interactive access to LLM-generated threat reports.

Another promising direction involves extending KLAGÉ to improve detection of zero-day attacks and unexpected behavioral triggers. These scenarios, by nature, escape traditional detection due to their novelty and lack of signature data. To address this, KLAGÉ could incorporate adversarial training strategies [68] and the synthetic injection of anomalous patterns [69] into knowledge graphs to simulate zero-day behavior during training.

Additional research should focus on mitigating the risks posed by LLMs, which may generate false or non-existent facts, or produce overconfident yet misleading explanations, especially when operating on incomplete, noisy, or ambiguous data. To mitigate these limitations and improve the trustworthiness of LLM-driven outputs, our future work will explore the integration of complementary strategies, including confidence estimation mechanisms that quantify uncertainty in model outputs and can trigger an alert when confidence is low [70].

Finally, another key avenue involves the integration of autonomous AI agents capable of not only detecting threats but also responding to them in real time, while maintaining explainability. We also aim to explore KLAGÉ's applicability in edge computing scenarios, where low-latency and lightweight models are essential, and in multilingual or multimodal threat detection, broadening the system's deployment across diverse contexts. These developments will further strengthen its role in bridging the gap between high-performance AI systems and the need for transparent, trustworthy cybersecurity solutions.

6. Conclusion

In this paper, we introduced KLAGÉ, a comprehensive methodology designed to enhance network security through advanced log classification and automated threat reporting. KLAGÉ integrates knowledge graphs and large language models to address key challenges in cybersecurity, including the detection of complex attacks such as DDoS, ARP poisoning, and advanced persistent threats. By employing Graph-BERT, KLAGÉ models intricate relationships within network traffic, producing enriched representations that support accurate anomaly detection, threat classification, and report generation. To ensure interpretability, the system incorporates XAI techniques, providing explanations for each detection and enabling transparent, human-understandable insights. Experimental evaluations demonstrate that KLAGÉ significantly improves performance over existing methods. Our approach achieved a classification accuracy of 84.11 %, outperforming baseline models by more than 5 %. The integration of LIME explanations into the report generation module has proven effective for producing clear and actionable summaries, assisting network administrators in identifying and mitigating threats more efficiently. Additionally, comparative evaluations—including ablation analysis, automated LLM-based assessments, and expert reviews—highlight the value of structured knowledge and explainability in enhancing report quality and trustworthiness. To facilitate reproducibility and support further research, the main code components of the KLAGÉ framework have been made publicly available.¹

CRedit authorship contribution statement

Loris Belcastro: Writing – original draft, Visualization, Validation, Writing – review & editing, Methodology; **Carmine Carlucci:** Software, Methodology, Conceptualization, Data curation, Formal analysis; **Cristian Cosentino:** Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization, Writing – review & editing; **Pietro Liò:** Validation, Supervision; **Fabrizio Marozzo:** Writing – review & editing, Writing – original draft, Validation, Supervision, Formal analysis, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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¹ <https://github.com/SCALabUnical/KLAGÉ>.

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